BTS-III(R/S)-01-23-1458 Reg. No. 20521066

A

B.Tech. Degree III Semester Regular/Supplementary Examination January 2023

ME 19-205-0305 METALLURGY AND MATERIAL SCIENCE

(2019 *Scheme*)

Time: 3 Hours Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Study the crystal structure, crystal systems, crystallographic planes, crystal imperfections, solidification, and diffusion.

CO2: Understand the phase rule, phase diagrams, solid solutions, examples of binary solid solutions.

CO3: Learn about different heat treatment processes for steel and applications of ferrous and nonferrous alloys.

CO4: Understand different failure mechanisms.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	What is the importance of Miller indices? Sketch (122) plane and [310] direction in simple cubic structure.	3	L1	1	3
	(b)	Classify crystal imperfections. Differentiate between line dislocation and screw dislocation.	3	L3	1	2
	(c)	Differentiate between carburising and nitriding surface treatment processes.	3	L2	1	1
	(d)	State and explain Gibb's phase rule.	3	L2	2	2
	(e)	What is the difference between failure of a material by creep and by stress rupture?	3	L2	4	2
	(f)	Differentiate between ductile fracture and brittle fracture.	3	L2	3	1
	(g)	What do you mean by super alloys and state their uses?	3	L2	3	1
	(h)	Write short note on Biomaterials.	3	L1	4	2

PART B

		(4 × 12 = 48)				
II.	(a)	Find the miller indices of a plane that makes an intercept of 1on the a-axis and 2 on the b-axis and parallel to the c-axis.	6	L5	1	3
	(b)	Compute the Atomic Packing Factor (APF) for Body Centred Cubic (BCC) and Face Centred Cubic (FCC) crystal structures.	6	L5	1	3
		OR				
III.	(a)	What is interplanar spacing and calculate the interplanar spacing of (220) set of planes of BCC iron if the lattice parameter of iron is 0.2866nm?	6	L5	1	4
	(b)	Differentiate between homogeneous nucleation and heterogeneous nucleation and cite some typical applications.	6	L2	1	2
IV.		Draw the Fe-Fe_3C phase diagram and label the phases and briefly explain the various invariant reactions.	12	L2	2	2
		OR				

(P.T.O)

			Marks	BL	CO	PO
V.	(a)	Differentiate between hardness and hardenability of steel and explain a test procedure to measure the hardenability of steel.	6	L2	2	2
	(b)	Draw a continuous-cooling transformation diagram (CCT diagram) for eutectoid plain-carbon steel. How does it differ from eutectoid isothermal transformation diagram (TTT diagram) for plain-carbon steel?	6	L3	2	3
VI.	(a)	Explain the difference between cold working and hot working.	6	L2	3	2
	(b)	Explain Griffith's theory of brittle fracture.	6	L3	3	3
		OR				
VII.	(a)	Explain elastic, anelastic and viscoelastic behaviour of materials and give examples.	6	L2	3	2
	(b)	What are strengthening mechanisms and explain any two strengthening mechanisms?	6	L2	4	2
VIII.	(a)	Write down the applications of ferrous and non-ferrous alloy steel.	6	L3	4	2
	(b)	What do you mean by stainless steel and explain the different types of it?	6	L2	4	2
		OR				
IX.	(a)	Explain with one example: (i) Shape memory alloys (ii) Aluminium alloys.	6	L2	4	2
	(b)	Write short notes on the following: (i) Titanium alloys and its use (ii) Optical fibres.	6	L2	4	2

Bloom's Taxonomy Levels

L1 = 7.5%, L2 = 55%, L3 = 15%, L4 = 2.5% , L5 = 15%, L6 = 5%

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